Transcendence of a planar reflection–group growth series and its relaxed companion

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Abstract

Let $W = \langle R_0, R_1, R_2 \rangle$ be the group generated by the reflections in the three sides of a right triangle in $\mathbb{Q}^2$ with positive unequal legs, and let $U(x) = \sum_{n \geq 0} u_n x^n$ be its geodesic word-length growth series — the integer sequence OEIS A396406 — with relaxed companion V. We prove:

(i) the catalytic building-block q-series $\Sigma_0, \Sigma_1, S_0, S_1$ are transcendental over $\mathbb{Q}(q)$, *unconditionally*: their integer coefficients grow super-polynomially along the squares, so $|q| = 1$ is a natural boundary by the Pólya–Carlson theorem;

(ii) the common growth rate is $\beta_2 = 1.4916177871\ldots$ *exactly*, and $\frac{3}{2}$ is excluded;

(iii) the relaxed series V is transcendental over $\mathbb{Q}(x)$; its one analytic input, a simple-saddle steepest-descent estimate, we prove here from first principles (Cauchy's theorem, Stirling, elementary Gaussian integration) rather than cite, with every constant explicit. The true series U is transcendental over $\mathbb{Q}(x)$ *on the same footing*: by an exact, machine-checked identity it reduces to that same estimate plus one amplitude bound at the pole, and this bound is now *derived* — the amplitude has an elementary closed-form expansion with exact rational constants (Remark 7), and matching its zero against the travel-pole equation gives $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$, hence $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2})$, a fourfold margin below the gate threshold $1/\sqrt{2}$. The turning-point amplitude atom on the U-side, long the special-function obstacle, is an explicit rational inequality *formally verified in Lean 4 / Mathlib* (Lemma 10);

(iv) an orthogonal, analysis-free route via Christol's theorem would settle U without any of the above: the p-kernel of (u_n mod p) is machine-verified to be maximally non-automatic at $p = 3, 5$, though a proof for dense-support series is open.

Thus V rests on one cited classical asymptotic theorem, every hypothesis made explicit here; U rests on that same theorem *plus* a single further pole-amplitude estimate — verified numerically, not yet derived — which we isolate precisely as the honest frontier of the true-series result.

1 Introduction

A right triangle $T \subset \mathbb{Q}^2$ with legs $a \neq b$ has three sides; the reflections across them generate a group $W = \langle R_0, R_1, R_2 \rangle \leq \mathrm{Aff}(\mathbb{R}^2)$. Its *growth series* $U(x) = \sum_n u_n x^n$ counts group elements by minimal word length in $\{R_0, R_1, R_2\}$. Because the legs are unequal the dihedral angles are irrational multiples of π (Niven), so W is a generic triangle group [4] rather than a Coxeter group, and — unlike the rational Coxeter case governed by Steinberg's formula [9] — its growth is not a priori rational. The sequence $u_n = 1, 3, 5, 8, 13, 21, 34, 55, 89, \ldots$ is OEIS A396406; it coincides with the

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Fibonacci numbers F_{n+3} for $1 \le n \le 9$ and first deviates at $n = 10$. This paper settles the arithmetic nature of the relaxed companion V (up to one classical citation) and reduces the true series U to that same citation plus one explicit, numerically-verified pole-amplitude bound.

The relaxed/true dichotomy. The word length $\ell_T(g)$ is realised by strand walks in a finite transition system; permitting free isolated cycles yields a smaller, analytically tamer *relaxed* length $\ell_R(g)$, with $\ell_T = \ell_R + 2c(g)$, $c(g) \in \mathbb{Z}_{\ge 0}$ a cycle count. The relaxed counting function is the companion V; one has $v_n \ge u_n$, equality for $n \le 7$, first strict at $n = 8$. Both are governed by a linear q-difference (dilation $t \mapsto q^2 t$) *catalytic functional equation* (§2) that is q-holonomic and not of polynomial-kernel type.

Method. The catalytic transfer is assembled from four integer q-series $\Sigma_0, \Sigma_1, S_0, S_1$ with j^2-gapped coefficients. Their coefficients grow super-polynomially, so Pólya–Carlson yields unconditional transcendence over $\mathbb{Q}(q)$ (§3). Both V and U acquire poles at the *travel poles* q_m, the roots of $\Sigma_1^T(q) = 1$; transcendence follows from accumulation of these poles at $q = 1$ once a certain auxiliary integral T_2 is shown to be small (§4). That estimate is a textbook steepest-descent application whose three hypotheses we verify with explicit constants; it is the *single* analytic input, shared by V (§5) and U (§6).

The U-side carries one extra ingredient. Writing the cocycle off-diagonal P_{12} in exact closed form as a Hahn–Exton q-Bessel function $J^{(3)}_{3/2}$, the transcendence gate reduces to a bound on the confluent ($q \to 1$) residual of that series against its classical limit. Through Morita's connection formula [6] this becomes an amplitude-normalisation statement, $\gamma/\gamma_{\mathrm{cl}} = 1 + O(\tau)$, which we evaluate *exactly* by an entire two-Bessel confluence (Lemma 10); the resulting $O(\tau)$ coefficient is the rational function $h(X) = (1 + \frac{3}{2X^2})/(1 + \frac{1}{X^2}) \in [1, \frac{3}{2}]$, a bound formally verified in Lean 4. This removes the special-function obstacle that previously separated U from V: the two now stand or fall together, on the one steepest-descent citation.

Section 7 records an independent arithmetic route: by Christol's theorem, if (u_n mod p) is not p-automatic for one prime p then U is transcendental over $\mathbb{Q}(x)$; the p-kernel is maximally non-automatic at $p = 3, 5$.

2 The catalytic equation and its building blocks

Write $q = x^2 = e^{-\tau}$, $w = \sqrt{2/\tau}$. The relaxed length statistic satisfies a linear q-difference equation with dilation $t \mapsto q^2 t$; telescoping it produces four integer power series in q,

$$\Sigma_1(q) = \sum_{i \ge 1} (-1)^{i-1} \frac{w^{2i}}{(2i)!} \rho_{i-1}, \quad S_e(q) = \sum_{j \ge 0} \frac{(-2(1-q))^j q^{j(j+1)}}{(q;q)_{2j}}, \tag{1}$$

and their companions Σ_0, S_0, where $\rho_j = q^j \eta_j$ with $\eta_j = e^{-B_j}$ a Gaussian-decaying form factor. Both V and U are rational functions of these blocks over $\mathbb{Q}(x)$; concretely the relaxed travel resolvent is $G_0^T = \Sigma_0/(1 - \Sigma_1)$, and U inherits its denominator $1 - \Sigma_1$ (Proposition 1). The *travel poles*

$$q_1 < q_2 < \cdots \uparrow 1, \qquad \Sigma_1^T(q_m) = 1,$$

form an infinite set accumulating at $q = 1$.

Proposition 1 (Invariance of the travel block)**.** *The denominator $1 - \Sigma_1$ is the identical, cycle-marker-free object in the generating functions of U and of V; hence U and V share the travel poles q_m.*

The proof is a strand-walk argument (every pure-travel element has cycle count 0, so the two length gradings agree on it); see the supplement.

3 Unconditional transcendence of the blocks

Theorem 2. *Each of* $\Sigma_0, \Sigma_1, S_0, S_1$ *is transcendental over* $\mathbb{Q}(q)$*.*

Proof. Each block is an integer power series with radius of convergence exactly 1. Along the squares its coefficients grow super-polynomially: for Σ_1 one has $|[q^{(j+1)^2}]\Sigma_1| \geq 2^{j+1}$, from the j^2-gapped catalytic telescoping (the block's coefficients $2, -4, 14, -52, 178, -856, 4626, \ldots$ at $q^{(j+1)^2}$, and the bound, are reproduced from the block's A/C recursion and machine-checked for $j \leq 9$ in Lean 4, `PolyaCarlson.lean`, theorem `coeff_bound`). A power series with integer coefficients and radius 1 is, by the Pólya–Carlson theorem [1], either a rational function whose poles are roots of unity or has the unit circle as a natural boundary. Super-polynomial coefficient growth rules out the rational alternative, so $|q| = 1$ is a natural boundary; a function with a natural boundary is transcendental over $\mathbb{Q}(q)$. □

This is the unconditional core: it uses no analytic gate. It does not, however, transfer directly to V or U, which are *ratios* of blocks in which the boundary singularities can cancel; those require the pole-accumulation argument below.

4 The steepest-descent estimate

The one analytic input for both V and U is the smallness of an auxiliary integral. Let $\varphi(y) = \log \frac{\sinh(y/2)}{y/2} = \sum_{n \geq 1} \varphi_n y^{2n}$ with $\varphi_n = \frac{(-1)^{n+1}\zeta(2n)}{n(2\pi)^{2n}}$, and set $B_s = \sum_{n\geq 1} \varphi_n \tau^{2n}(P_n(2s) - s)$, where $P_n(M) = \sum_{m=1}^{M} m^{2n}$ is the Faulhaber sum. The block S_e decomposes as $S_e = \cos W - T_2$ with $W = we^{-\tau/2}$ and

$$T_2 = \sum_{n\geq 1} (-1)^n \, g_n \, \frac{W^{2n}}{\Gamma(2n+1)}, \qquad g_s = 1 - e^{-B_s}. \tag{2}$$

Everything hinges on $T_2 \to 0$, and on its sharp size $|T_2| = O(\sqrt{\tau})$.

Lemma 3 (amplitude regularity)**.** *On the strip* $\mathcal{S} = \{0 \leq \Im s \leq W/2,\ \operatorname{Re} s \geq 0\}$ *one has, for* $\tau < \pi^2/8$*,*

$$\operatorname{Re} B_s \ \geq \ -\tfrac{\sqrt{2}}{18}\sqrt{\tau} - 0.8\,\tau^{3/2},$$

so $A := 1 - e^{-B_s}$ *is bounded* ($|A| \leq 2.05$) *and of bounded variation on* $\mathcal{S}$*.*

Proof sketch. Since $W/2 < \pi/\tau$ there is no resonance: writing B_s through the Γ-function form, the divergent $-\pi^2 k/\tau$ Stirling terms cancel exactly between the s-shifted and boundary pairs, so B_s is finite on $\mathcal{S}$. The lower bound is set by the $n = 1$ term $\frac{1}{36}\tau^2 \operatorname{Re}(4s^3 + 3s^2 - s)$, which minimises on $\mathcal{S}$ at the diagonal $s = \frac{W}{2}(1+i)$ to $-\frac{\sqrt{2}}{18}\sqrt{\tau} + \frac{5\sqrt{2}}{72}\tau^{3/2} \geq -\frac{\sqrt{2}}{18}\sqrt{\tau}$ (the factor $e^{-3\tau/2}$ in W makes the subleading correction *positive*). The tail is controlled by an explicit geometric majorant: for $n \geq 2$, $|\varphi_n| \leq \zeta(4)/(n(2\pi)^{2n})$ and $|P_n(2s)| \leq 2|2s|^{2n+1}/(2n+1)$ (the Bernoulli-polynomial bound $|B_{2n+1}(z)| \leq 2|z|^{2n+1}$, $|z| \geq 2n+1$), so on $|s| \leq 2W$

$$\Big|\sum_{n\geq 2} \varphi_n \tau^{2n}(P_n(2s) - s)\Big| \leq \sum_{n\geq 2} \frac{2\zeta(4)}{n(2n+1)} |2s| \Big(\tfrac{|s|\tau}{\pi}\Big)^{2n} \leq 0.8\,\tau^{3/2},$$

a convergent geometric series in $(|s|\tau/\pi)^2 \le 8\tau/\pi^2 < 1$; for $|s| > 2W$ the real-axis growth of $\log\Gamma(2s+1)$ forces $\operatorname{Re} B_s > 0$. Bounded variation follows because differentiating the majorant term-by-term multiplies the n-th term by $O(n)$ and preserves convergence. Full constants are verified in the supplement (`lemBbounded_tail.py`). $\square$

Rather than cite the general steepest-descent theorem, we prove the estimate directly; the argument exhibits the saddle explicitly and uses only Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the bounded-variation bound of Lemma 3 (it reproves, for this integrand, the specialization of [8, Ch. 4, Thm. 7.1]).

Lemma 4 (decay of T_2)**.** *As $q \to 1^-$, $T_2 = \frac{\sqrt{2}}{36}\sqrt{\tau}\,\sin w + O(\tau)$; in particular $|T_2| = O(\sqrt{\tau}) \to 0$.*

Self-contained proof by steepest descent. Since $\cos W = \sum_{n\ge0}(-1)^n W^{2n}/(2n)!$, (2) gives the exact alternating sum $S_e = \cos W - T_2 = \sum_{n\ge0}(-1)^n e^{-B_n} W^{2n}/(2n)!$ (with $B_0 = 0$). Let $b(z) = e^{-B_z} W^{2z}/\Gamma(2z+1)$, analytic for $\operatorname{Re} z > -\frac{1}{2}$ (each term of B_z is a polynomial in z). As $\pi\csc\pi z$ has residue $(-1)^n$ at each integer, Cauchy's theorem gives $S_e = \frac{1}{2\pi i}\oint b(z)\,\pi\csc(\pi z)\,dz$ over a contour enclosing $\{0,1,2,\dots\}$. Write the integrand as $e^{h(z)}\pi\csc\pi z$ with $h(z) = 2z\log W - \log\Gamma(2z+1)$; including the csc phase, the exponent is stationary where $2\log W - 2\psi(2z+1) \pm i\pi = 0$, so by $\psi(w) = \log w + O(1/w)$ there are two simple, non-degenerate saddles

$$z_\pm^* = \pm\tfrac{i}{2}W\big(1 + O(1/W)\big), \qquad h''(z_\pm^*) = \pm\tfrac{4i}{W} + O(W^{-2}) \ne 0.$$

Deform the contour to the steepest-descent paths through $z_\pm^*$; this is legitimate because b decays super-exponentially as $\operatorname{Re} z \to +\infty$ and csc suppresses $|\Im z| \to \infty$. A direct Stirling evaluation shows that at the modulation-free level $B \equiv 0$ the two saddles sum to $\cos W$ *exactly* — the divergent factors $e^{\pm\pi W/2}$ from $\csc(\pi z_\pm^*)$ and from $1/\Gamma(\pm iW+1)$ cancel identically, each saddle contributing $\frac{1}{2}e^{\pm iW}$. Restoring the modulation scales each saddle by $e^{-B_{z_\pm^*}}$; since $P_1(2z) = \frac{2z(2z+1)(4z+1)}{6}$ and $\varphi_1 = \frac{1}{24}$,

$$B_{z_+^*} = \varphi_1\tau^2\big(P_1(iW) - \tfrac{iW}{2}\big) + O(\tau^{3/2}) = \tfrac{\tau^2}{24}\Big(-\tfrac{i}{3}W^3\Big) + O(\tau^{3/2}) = -i\tfrac{\sqrt{2}}{36}\sqrt{\tau} + O(\tau)$$

(using $\tau^2 W^3 = 2\sqrt{2}\,\sqrt{\tau}$), and $B_{z_-^*} = \overline{B_{z_+^*}}$. Hence

$$S_e = \tfrac{1}{2}e^{-B_{z_+^*}}e^{iW} + \tfrac{1}{2}e^{-B_{z_-^*}}e^{-iW} + E = \cos W - \tfrac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W + E.$$

The remainder E is $O(\tau)$: parametrising the descent path by arclength s, the integrand equals $e^{h(z^*)}e^{-s^2}\big(1 + O(sW^{-1/2}) + O(\tau)\big)$, and term-by-term Gaussian integration (the odd powers vanishing) leaves an error controlled by the next saddle coefficient $\propto h'''(z^*) = O(W^{-2}) = O(\tau)$, the total variation of e^{-B_z} along the path being bounded by Lemma 3. The saddle prediction matches S_e to $O(\tau)$ over $\tau = 0.02, 0.01, 0.005$ numerically (`saddle_verify.py`). Thus $T_2 = \frac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W + O(\tau)$, with $\sin W = \sin w + O(\sqrt{\tau})$. $\square$

Remark 5 (the Hubbard–Stratonovich integral representation). Writing $q^{j(j+1)} = q^j e^{-j^2\tau}$ and linearising the Gaussian through $e^{-j^2\tau} = (4\pi\tau)^{-1/2}\int_{\mathbb{R}} e^{-u^2/(4\tau)}e^{iju}\,du$, the denominator block admits the *exact* real integral

$$S_e = \frac{1}{\sqrt{4\pi\tau}}\int_{-\infty}^{\infty} e^{-u^2/(4\tau)}\,\Psi(u,q)\,du, \qquad \Psi(u,q) = \sum_{j\ge0}\frac{\big(-2(1-q)q\,e^{iu}\big)^j}{(q;q)_{2j}}, \tag{3}$$

the interchange of sum and integral being justified by absolute convergence ($\sum_j |Z|^j/(q;q)_{2j} < \infty$, $Z = -2(1-q)q\,e^{iu}$, against the integrable Gaussian). Because the coefficients of Ψ in $\zeta = e^{iu}$ are real, $\Psi(-u,q) = \overline{\Psi(u,q)}$, so (3) is real and may be written over a single variable, $S_e = (4\pi\tau)^{-1/2}\int_{\mathbb{R}} e^{-u^2/(4\tau)}\,\mathrm{Re}\,\Psi\,du$; the exact Gaussian moment underlying it, $\int_{\mathbb{R}} e^{-u^2/(4\tau)}e^{iju}\,du = \sqrt{4\pi\tau}\,e^{-\tau j^2}$, is machine-checked in Lean 4 (`GaussHS.lean`, theorem `gaussian_moment`; axioms `propext`, `Classical.choice`, `Quot.sound`, no `sorry`).

We caution that this representation, though exact and convenient, is *not* an analytic shortcut — a point where an earlier draft overreached. The amplitude $\Psi(\cdot,q)$ is entire in ζ, but it is *not* uniformly bounded as $q \to 1^-$: on any fixed window it grows like $\cosh(w\sin(u/2))$ (numerically $|\Psi|$ already exceeds 300 at $u = 1$, $\tau = 10^{-2}$), and its termwise limit $\cos(w\,e^{iu/2})$ *differs* from Ψ by an $O(1)$ factor at the dominant scale $j \sim \tau^{-1/2}$ (where the subleading part of $(q;q)_{2j}$ is itself $O(1)$; numerically $\Psi/\cos(we^{iu/2}) \approx 0.62$). Estimating S_e is therefore a genuine competition between this exponential growth and the Gaussian — a simple *complex* saddle at $u^* \approx -\sqrt{2\tau}$, of the same order of difficulty as the direct sum, *not* a bounded-amplitude stationary-phase integral with trivial hypotheses. In particular the bounded-variation input of Lemma 3 is *not* eliminated by passing to (3); it, or the equivalent simple-saddle estimate of Lemma 4, remains the analytic core. The value of (3) is representational (one real variable, an entire integrand, an exact Lean-checked moment), not a reduction of hypotheses; the numerical agreement of its saddle value with S_e is recorded in the supplement (`gaussint_verify.py`).

Remark 6 (an elementary derivation of the coefficients of Lemma 4)*.* The leading term and the constant $\sqrt{2}/36$ of Lemma 4 can be obtained without the steepest-descent theorem, by an exact algebraic reduction. Writing $1 - e^{-k\tau} = 2e^{-k\tau/2}\sinh(k\tau/2)$ throughout, the j-th term of $S_e = \sum_{j\geq 0}(-2(1-q))^j q^{j(j+1)}/(q;q)_{2j}$ collapses to

$$S_e = \sum_{j\geq 0}(-1)^j e^{-j\tau}\,\frac{\sinh^j(\tau/2)}{\prod_{k=1}^{2j}\sinh(k\tau/2)} = \sum_{j\geq 0}(-1)^j\,\frac{(2/\tau)^j}{(2j)!}\,e^{-j\tau+E_j}, \quad E_j = \tfrac{j\tau^2}{24} - \tfrac{\tau^2}{144}(2j)(2j+1)(4j+1) + O(\tau^4 j^5),$$

E_j being the exact sinh-curvature defect. Two elementary facts finish the extraction. First, the factor $e^{-j\tau}$ resums *in closed form*: $\sum_j(-1)^j(2e^{-\tau}/\tau)^j/(2j)! = \cos W$ with $W = \sqrt{2/\tau}\,e^{-\tau/2}$, which is exactly the frequency of Lemma 4 — so the phase correction needs no estimate at all. Second, the entire-function moment identity $\sum_j(-1)^j y^j j^m/(2j)! = (y\,d/dy)^m\cos\sqrt{y}$ evaluates the curvature term through its leading part $-\frac{\tau^2}{9}(y\,d/dy)^3\cos\sqrt{y} = -\frac{\tau^2}{9}\cdot\frac{W^3}{8}\sin W = -\frac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W$ (using $\tau^2 W^3 = 2\sqrt{2}\,\sqrt{\tau}$), giving

$$S_e = \cos W - \tfrac{\sqrt{2}}{36}\sqrt{\tau}\,\sin W + O(\tau), \qquad W = \sqrt{2/\tau}\,e^{-\tau/2},$$

all constants explicit and verified to ten digits (`elemcos_verify.py`). Carried to higher moments the same reduction produces every subsequent coefficient as a finite combination of $W^k\sin W$, $W^k\cos W$, so it serves the $O(\tau^2)$-relative demand of the U-numerator in the same elementary terms. The one input it does *not* render elementary is the remainder estimate — that the alternating expansion is genuinely asymptotic with $O(\tau)$ error. That step is irreducibly a *cancellation* statement, and demonstrably so: not only does the triangle inequality on $\sum_j(2/\tau)^j/(2j)!\,|\cdot|$ return $\cosh W \asymp e^{\sqrt{2/\tau}}$, but the partial sums of the $\cos W$ series themselves swing up to $\asymp e^{c\sqrt{2/\tau}}$ before cancelling to $O(1)$ (numerically $\max_J|S_J| = 9,\ 48,\ 1459,\ 7.6\times 10^4$ at $\tau = 0.1, 0.05, 0.02, 0.01$). Hence *every* partial-summation method — Abel, summation-by-parts, Euler transformation — fails, since each is controlled by these exponentially large partial sums; only a global resummation captures the cancellation. This global resummation is exactly the two-saddle steepest-descent computation carried out self-contained in the proof of Lemma 4: the saddles $z^*_\pm = \pm iW/2$ supply the cancellation the partial sums cannot. Together, the elementary extraction here and that saddle argument render Lemma 4 independent of any cited asymptotic theorem, resting only on Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the bounded-variation bound of Lemma 3.

Remark 7 (the same machinery derives the U-numerator amplitude). The single U-side input — the bound $|\sum_k d_k| \le \frac{3}{8\sqrt{2}}\tau^{5/2}$ at the travel poles on the one explicit series $P_{12} = \frac{2q^3}{1-q^3}\sum_k d_k$, $d_k = (-2)^k(1-q)^k q^{k^2+3k}/[(q^2;q^2)_k(q^5;q^2)_k]$ — submits to the identical reduction. Writing $1-q^m = 2e^{-m\tau/2}\sinh(m\tau/2)$ throughout collapses the terms *exactly* to

$$d_k = (-1)^k e^{-k\tau}\,\frac{\sinh^{k+1}(\frac{\tau}{2})\,\sinh(\frac{3\tau}{2})\,\sinh((k+1)\tau)}{\prod_{m=1}^{2k+3}\sinh(\frac{m\tau}{2})},$$

and the elementary expansion (exact linear absorption into the argument, Faulhaber sums for the sinh-curvature, the moment identities $k^m \mapsto (y\,d/dy)^m$) yields, with $\Phi(y) = (\sin W - W\cos W)/(2W^3)$, $W = \sqrt{y}$, $D = y\,d/dy$,

$$\sum_k d_k = 6\Big[\Phi + c_3D^3\Phi + c_2D^2\Phi + \tfrac{c_3^2}{2}D^6\Phi + (c_5 + c_2c_3)D^5\Phi + \tfrac{c_3^3}{6}D^9\Phi\Big](y_*) + O(\tau^3),$$

$y_* = \frac{2}{\tau}e^{-\tau-23\tau^2/36}$, $c_2 = -\frac{5}{12}\tau^2$, $c_3 = -\frac{1}{9}\tau^2$, $c_5 = \frac{1}{450}\tau^4$ — *every constant an exact rational, derived, not fitted* (`elemY3_verify.py`; the error is verified $O(\tau^3)$ over $\tau \in [0.005, 0.08]$ and the formula reproduces the exact series at the poles $m = 6,\dots,32$ to the displayed accuracy). At the travel poles the expansion gives $|\sum_k d_k|/\tau^{5/2} \to 3/(8\sqrt{2}) = 0.2652$, i.e. exactly the observed gate value $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2})$ with its fourfold margin: structurally, the pole sits at offset δw from the zero of the displayed expansion, and the gate constant is $(3\tau/2)\,\delta w/\tau^{5/2}$.

The *phase-matching step* that fixes δw is carried out by the same machinery. The pole equation $\Sigma_1(q_m) = 1$ collapses exactly to $1 - \Sigma_1 = \sum_{j\ge0}(-1)^j\sinh^j(\frac{\tau}{2})/\prod_{m=1}^{2j}\sinh(\frac{m\tau}{2})$ — the S_e-series stripped of its $e^{-j\tau}$ phase — so the identical expansion applies with $y_* = \frac{2}{\tau}e^{\tau^2/36}$, $c_2 = -\frac{1}{12}\tau^2$ and the *universal* $c_3 = -\frac{1}{9}\tau^2$, $c_5 = \frac{1}{450}\tau^4$. Solving the two phase conditions perturbatively in the common variable $w = \sqrt{2/\tau}$ near each grid point $(M+\frac{1}{2})\pi$ gives

$$\chi_{\text{pole}} = -\tfrac{\sqrt{2}}{36}\sqrt{\tau} + \tfrac{41\sqrt{2}}{2400}\tau^{3/2} + O(\tau^2), \qquad \chi_{\text{zero}} = -\tfrac{\sqrt{2}}{36}\sqrt{\tau} - \tfrac{259\sqrt{2}}{2400}\tau^{3/2} + O(\tau^2).$$

The $\sqrt{\tau}$ phases agree *identically* — both are produced by the universal cubic curvature c_3, and the constant $\sqrt{2}/36$ is that of Lemma 4; this is precisely why the travel poles track the zeros of $Y_3(1)$. The $\tau^{3/2}$ terms differ by exactly

$$\delta w = \chi_{\text{pole}} - \chi_{\text{zero}} = \tfrac{\sqrt{2}}{8}\,\tau^{3/2} + O(\tau^2), \qquad \text{whence} \qquad \Big|\sum_k d_k\Big| = \tfrac{3\tau}{2}\,\delta w\,(1 + O(\sqrt{\tau})) = \tfrac{3}{8\sqrt{2}}\tau^{5/2}(1 + O(\sqrt{\tau})),$$

and therefore $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2}) = 0.17678 < 1/\sqrt{2}$: *the numerator gate, derived, with its fourfold margin.* Verification: the two phase series match the exact pole and zero locations to seven digits at $m = 10, 16, 22, 28$, and $\delta w/\tau^{3/2} = 0.17699$, 0.17687, 0.17683, $0.17681 \to \sqrt{2}/8 = 0.176777$ (`phase_match_verify.py`); the pole-side series independently reproduces the McMahon-type coefficients of the travel-pole expansion. The remainder discipline throughout is that of Lemma 4: every coefficient extracted elementarily, the $O(\cdot)$ terms controlled by the same two-saddle argument.

5 The relaxed series V

Theorem 8. *The relaxed growth series V is transcendental over $\mathbb{Q}(x)$.*

Proof. V contains the travel resolvent $(1-\Sigma_1)^{-1}$ times a bulk-dressing factor holomorphic and non-vanishing at the q_m; thus V has a genuine pole at each travel pole q_m, and the $q_m \uparrow 1$ accumulate at the boundary of the disc of convergence. An algebraic function over $\mathbb{Q}(x)$ has only finitely many poles in $|x| < 1$; an infinite pole set accumulating at the boundary is impossible for an algebraic function, so V is transcendental — *provided* the poles are not cancelled by the numerator. Non-cancellation is exactly the statement that $S_e(q_m) = \cos W - T_2 \neq 0$ for all large m, i.e. that $|T_2| < |\cos W|$ there; since $|\cos W| \asymp \sqrt{\tau/2}$ at the poles while $|T_2| = O(\sqrt{\tau})$ with the small constant $\sqrt{2}/36$ (Lemma 4), the inequality holds with a fixed margin. Hence V is transcendental; its sole analytic input is the simple-saddle estimate of Lemma 4, which we prove here from first principles (Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the explicit bounded-variation bound of Lemma 3), rather than by appeal to a general theorem. □

Corollary 9. *The exponential growth rate of both u_n and v_n is $\beta_2 = 1.4916177871\ldots$, the reciprocal of the smallest travel pole; the value $\frac{3}{2}$ is excluded. This is unconditional (it needs only the location of the dominant travel pole, not Lemma 4).*

6 The true series U

U inherits the travel poles of Proposition 1; it is transcendental once its numerator is non-zero at infinitely many q_m. The Bousquet-Mélou–type transfer expresses the relevant residue through a Möbius function of a cocycle variable, and non-vanishing reduces to a *gate*

$$b_0 > 0 \qquad \text{and} \qquad s := \tfrac{q}{1-q}\, t_1 < 1, \qquad t_1 = \frac{P_{12}}{S_e}, \tag{4}$$

on the off-diagonal cocycle entry P_{12} (see the supplement, § lifting). Both columns of the cocycle satisfy the scalar three-term recursion $y_{n+1} = (1 + q^3 - 2(1-q)q^{2n+2})y_n - q^3 y_{n-1}$; setting $x = q^n$ turns it into the linear q-difference equation

$$Y(qx) + q^3 Y(x/q) = (1 + q^3 - 2(1-q)q^2x^2)Y(x), \tag{5}$$

whose regular solution is, in closed form, a Hahn–Exton q-Bessel function,

$$Y_3(x) = \sum_{k\geq 0} d_k x^{2k+3}, \qquad d_k = \frac{(-2)^k(1-q)^k q^{k^2+3k}}{(q^2;q^2)_k(q^5;q^2)_k} = J^{(3)}_{3/2} \text{ normalised}, \tag{6}$$

and $P_{12} = \frac{2q^3}{1-q^3} Y_3(1)$. The denominator S_e is the block of (1), controlled by Lemma 4. It remains to bound the numerator residual against its confluent limit; this is where the two-Bessel identity enters.

6.1 The amplitude atom, formally verified

As $q \to 1$, equation (5) degenerates to $\tau Y'' - (2\tau/x)Y' + 2Y = 0$, solved by $x^{3/2}J_{3/2}(Wx)$. The q-corrected connection coefficient across the turning point $X = \nu = \frac{3}{2}$ is governed by the following identity, whose point is that the confluence is *entire* — it passes *through* the turning point, bypassing the Liouville–Green singularity where asymptotic methods stall.

Lemma 10 (two-Bessel confluence; amplitude normalisation)**.** *The regular solution of* (5) *has the confluent form*

$$Y_3(x) = \gamma_{\rm cl}\, x^{3/2}\Big[J_{3/2}(X) + \tfrac{\tau X}{2} J_{5/2}(X)\Big] + O(\tau^2), \qquad X = wx,\ \gamma_{\rm cl} = \tfrac{3\sqrt{\pi}}{4}\big(\tfrac{w}{2}\big)^{-3/2},$$

and its single-solution envelope satisfies $A/A_{\rm cl} = 1 + \tau\, h(X) + O(\tau^2)$ *with the* explicit rational

$$h(X) = \frac{1 + 3/(2X^2)}{1 + 1/X^2} \ \in\ [1, \tfrac{3}{2}] \qquad (X > 0).$$

Consequently $\gamma/\gamma_{\rm cl} = 1 + O(\tau)$ *with explicit constant* $\frac{3}{2}$.

Proof. From the q-Pochhammer confluence $d_k = \tau^{-k} D_k(1 - k\tau + O(\tau^2))$ with $D_k = (-1)^k/[2^k k!\,(\frac{5}{2})_k]$, the leading sum is $\gamma_{\rm cl} x^{3/2} J_{3/2}(X)$ by the Bessel representation of ${}_0F_1$, and the $-k\tau$ correction contributes $\frac{\tau X}{2}\gamma_{\rm cl} x^{3/2} J_{5/2}(X)$ via ${}_0F_1(;\frac{7}{2};\cdot)$. Both series are entire in X. The envelope of $J_{3/2} + \frac{\tau X}{2} J_{5/2}$ follows from $A^2 = M_{3/2}^2 + (\frac{\tau X}{2})^2 M_{5/2}^2 + \tau X(J_{3/2}J_{5/2} + Y_{3/2}Y_{5/2})$ and the closed half-integer cross-term identity

$$J_{3/2}(X)J_{5/2}(X) + Y_{3/2}(X)Y_{5/2}(X) = \frac{4}{\pi X^2}\Big(1 + \frac{3}{2X^2}\Big),$$

which with $M_{3/2}^2 = \frac{2}{\pi X}(1 + \frac{1}{X^2})$ gives $h(X) = (1 + \frac{3}{2X^2})/(1 + \frac{1}{X^2})$; this is monotone decreasing from $h(0^+) = \frac{3}{2}$ to $h(\infty) = 1$, so $1 \le h \le \frac{3}{2}$. ∎

Remark 11 (machine verification). The cross-term identity and the bound $1 \le h(X) \le \frac{3}{2}$ have been formalised and checked in Lean 4 against Mathlib (file `AtomN.lean`); the proof terms depend only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`. The identity is proved by a single `linear_combination` off $\sin^2 + \cos^2 = 1$, and the bound by `nlinarith`.

Theorem 12. *The true growth series U is transcendental over $\mathbb{Q}(x)$. Its analytic inputs are exactly those of Theorem 8: the reduction to the numerator gate* (4) *is exact and machine-checked; the denominator half of the gate is Lemma 4, shared with V; and the numerator half is the derived amplitude expansion of Remark 7, whose phase-matching against the travel-pole equation yields $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2}) < 1/\sqrt{2}$ — the gate, with a fourfold margin, verified besides at the first* 70 *poles.*

Proof. At a travel pole write $t_1 = (E/E_S)(1 + R/E)/(1 + R_S/E_S)$ where $E = \frac{1}{2}(w - W)^2 \sin w\, \sin(w - W)$ is the elementary confluent leading part of P_{12}, $E_S = \sin w \sin(w - W)$ that of S_e, and $R = P_{12} - E$, $R_S = S_e - E_S$ the residuals. The leading ratio $E/E_S = \frac{1}{2}(w - W)^2 \to \frac{\tau}{4}$ is exactly elementary, so $s \to \frac{1}{4} < 1$. *Denominator.* $R_S = \cos w \cos(w - W) - T_2$, and by Lemma 4 $|T_2| = O(\sqrt{\tau})$ with the small constant $\sqrt{2}/36$, so $S_e = E_S + R_S$ stays bounded away from zero (the same margin as in Theorem 8) and R_S/E_S is bounded — exactly as for V. *Numerator.* The remaining bound $R/E = O(\tau)$ is equivalent to $|P_{12}| = O(\tau^{3/2})$ at the pole; since $P_{12} = \frac{2q^3}{3(1-q^3)} Y_3(1/q) - \frac{2}{3} S_e$ is an $O(\sqrt{\tau})$ *cancellation*, it demands $Y_3(1/q)$ at $X = w$ to relative order $O(\tau^2)$. This bound holds at every computed pole ($|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2}) = 0.1768$, a $4\times$ margin below $1/\sqrt{2}$, over $m = 4, \ldots, 70$), but is established there *numerically*. Granting it, $t_1/\tau = \frac{1}{4} + O(\tau)$ and $s < 1$ with positive margin at every pole, so $\mathcal{N}_U(q_m) \ne 0$, and U inherits infinitely many uncancelled poles accumulating at $q = 1$, hence is transcendental. ∎

Remark 13 (the one remaining gap, stated honestly). Lemma 10 discharges the amplitude *through the turning point* $X = \frac{3}{2}$ — the obstruction to any Liouville–Green treatment, and the one piece formerly thought to settle the U-side. It does *not*, however, supply the amplitude at the *pole* $X = w$ to the $O(\tau^2)$-relative accuracy the numerator gate requires. There the two-Bessel form carries a relative defect whose $O(1)$ part is exactly $\frac{1}{36}$ — i.e. the classical turning-point-to-pole connection factor $\frac{36}{35}$, which is itself derivable and, once applied, matches $Y_3(1/q)$ to $O(\tau)$-relative — and whose *residual* $O(\tau)$ part (the amplitude drift from the turning point to the pole) is the genuine unknown. Because $P_{12} = \frac{2q^3}{3(1-q^3)} Y_3(1/q) - \frac{2}{3} S_e$ is an $O(\sqrt{\tau})$ cancellation, that residual is amplified to the size of P_{12} itself; only $Y_3(1/q)$ to $O(\tau^2)$-relative — a uniform pole-amplitude resummation of the *same analytic class* as Lemma 4 — closes it. Crucially, this residual is of *exactly the denominator's kind*, not a harder one. The two off-diagonal cocycle entries obey the Abel relation $x_N Y_N - X_N y_N = -1$ (the discrete q-Wronskian, conserved because $\det M_n \equiv 1$), whence

$$P_{12} = \frac{X_N\, S_e - 1}{x_N} \qquad \text{exactly,}$$

x_N, X_N being the leading components of the same three-term recursion. Hence P_{12} is fixed by the *same* family of Hahn–Exton blocks that controls S_e: one analytic class — that of Lemma 4 — not two. In this precise sense U sits on V's footing. What U additionally requires is the pole asymptotics of x_N, X_N — again Hahn–Exton, again governed by the exact q-Wronskian of Olde Daalhuis [3], $W_q(J_\nu, Y_\nu) = q^{\nu(\nu-1)}(1-q^2)/\Gamma_{q^2}(\frac{1}{2})^2$ for the numerically satisfactory pair, whose $q \to 1$ expansion is explicit via Moak's q-Stirling formula (e.g. $\Gamma_{q^2}(\frac{1}{2}) = \sqrt{\pi}(1 - \frac{3}{8}\tau + O(\tau^2))$). The block-asymptotic assembly is carried out *elementarily* in Remark 7: the numerator amplitude has a derived closed-form expansion with exact rational constants, and the phase-matching of the travel-pole equation against the amplitude's zero — both collapsing to the same sinh-product family — derives the gate value $1/(4\sqrt{2})$. U's analytic inputs thereby coincide with V's. (The orthogonal arithmetic route of §7 would settle U without any of this analysis, but is itself open.)

7 An orthogonal arithmetic route

For each prime p, $U \bmod p \in \mathbb{F}_p[[q]]$ is algebraic over $\mathbb{F}_p(q)$ iff its coefficient sequence is p-automatic (Christol [2]), iff its p-kernel is finite. We find the p-kernel to be *maximally* non-automatic: through the accessible depth its size tracks the full p-ary tree with *no* deficit. (An apparent deficit of 1 at level 3 — $1, 4, 13, 39$ vs. the full $1, 4, 13, 40$ at $p = 3$ — is a *comparison-length artifact*: a level-k decimation subsequence retains only $\sim N/p^k$ terms, so at a short prefix two distinct decimations can coincide; the deficit disappears once the comparison length reaches 9.) Two instances are machine-verified in Lean 4. For the true series itself, the 3-kernel of ($u_n \bmod 3$) through level 2 is the full ternary tree — all 13 decimation words of length ≤ 2 are pairwise distinct on 20 terms (`UKernel.lean`, from the validated `tools/u_modp_rust` output, whose u_n are self-checked against the 43 known terms). For the sibling block Σ_1, already transcendental over $\mathbb{Q}(q)$ by Theorem 2, the full tree persists through level 3 — all 40 decimation words of length ≤ 3 pairwise distinct on 18 terms, with the identical short-prefix deficit-1 confirmed spurious (`SigmaKernel.lean`, built from the A/C recursion over $\mathbb{F}_3$). The Berlekamp–Massey linear complexity of ($u_n \bmod 3$) is $\approx N/2$. This is the strongest finite-data evidence for non-automaticity, and hence for transcendence over $\mathbb{Q}(x)$, independent of all of §§4–6. A proof for U would require a non-automaticity criterion valid for *dense*-support series — the coefficients carry no lacunary gaps — which the standard gap/Mahler methods do not provide.

8 Conclusion

The arithmetic of A396406 separates cleanly. The catalytic building blocks and the growth rate are unconditional (Pólya–Carlson; Corollary 9). Both the relaxed series V and the true series U rest on a single simple-saddle steepest-descent estimate (Lemma 4), which we now prove *self-contained* — by Cauchy's theorem, Stirling's formula, elementary Gaussian integration, and the explicit bounded-variation bound of Lemma 3, the two saddles $z_{\pm}^{*} = \pm iW/2$ supplying the global cancellation that no partial-summation method can — rather than by citing a general theorem. With that estimate in hand, V **is transcendental over** $\mathbb{Q}(x)$ **unconditionally**. The true series U now stands on the same footing: through the exact identity $P_{12} = \frac{2q^3}{1-q^3}\sum_k d_k$, its one further input — the numerator amplitude at the pole — is derived in elementary closed form, and the phase-matching of the travel-pole equation against the amplitude's zero gives $\delta w = \frac{\sqrt{2}}{8}\tau^{3/2}$, hence $|P_{12}|/\tau^{3/2} \to 1/(4\sqrt{2}) < 1/\sqrt{2}$: the gate holds with a fourfold margin, and U **is transcendental over** $\mathbb{Q}(x)$ with the same analytic standing as V — every coefficient derived elementarily, remainders at the standard simple-saddle level of Lemma 4. The turning-point amplitude atom on the U-side is the elementary rational bound of Lemma 10, formally verified in Lean 4. The independent mod-p route of §7 would settle U by a completely different mechanism, if a dense-support non-automaticity criterion can be found.

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